

Litter recognition

Moss&Gund - business

Proposal

One of the largest problems on our major roads is the collection of litter and detritus at the side of the road.

Due to increased usage of non-biodegradable materials in packaging, increased propensity to dispose of litter from moving vehicles, and increased catastrophic deterioration of vehicle components (plastic undertrays, tyres etc), litter levels are reaching a level dangerous to the local environment, water courses and even traffic.

Added on to this is the removal of the 'hard shoulder' lane from smart motorways, which restricts access to the verge at the side of the road, and requires expensive lane closure and a greater risk to workers manually collecting the litter.

What is needed is a system to autonomously detect and dispose of roadside litter, which often collects in the rough grassland, shrubs and trees next to the road.

This proposal is for a system to

1. Detect the presence of man made items in the verge and roadside - on the ground and in the trees and shrubs.
2. Differentiate between different sizes and categories of litter
3. Determine methods of effective collection of the litter depending on size and category
4. Place the collected litter into a container for storage.

This system should be ultimately be designed to work in the required roadside environment, and consideration should be made for its placement on a small autonomous vehicle.

Information for students

This is an ambitious project and you will be creating a proof of concept. The full brief will not be possible to complete - you will need to work with the client to agree an achievable goal for the project.

This will suit ambitious students already interested in computer vision and robotics, who are prepared to do research into state-of-the-art, commercially available solutions.

Why the client is interested in working with you

This challenge, while novel, allows for the use of known technologies employed in a focused task, which I believe is an ideal challenge for students. It can be done using low cost equipment which I believe would be available to the University. It is a highly marketable project which would provide a terrific case study for students to demonstrate their skills for future employers. This project is aimed at becoming a commercial solution in the next 2-3 years, and as such could provide direct follow on job opportunities for successful students.